Hulki Çıray

hulkiciray@gmail.com | +90 534 528 65 14 | Portfolio | LinkedIn | GitHub

Education

Koç University — Istanbul, Turkey
Master of Science in Computer Science and Engineering | CGPA: 3.29 — *2023 – 2026*
Thesis: SWIFT: Selective Weight Identification and Fine Tuning | (Link)
Advisor: Prof. Gözde Gül Şahin

Istanbul Technical University — Istanbul, Turkey
Bachelor of Science in Electrical Engineering | CGPA: 3.02 — *2016 – 2021*
Graduation Project: Object Detection with Deep Learning for Autonomous Cars
Advisor: Prof. Tahir Çetin Akıncı

Izmir Ataturk High School — Izmir, Turkey
Math-Science Track — *2011 – 2015*

Research Experience

AI Researcher — Feb 2025 – Present
TÜBİTAK - Koç University — *Istanbul, Turkey*
- Developed **SWIFT**, a novel Selective Parameter Efficient Fine Tuning framework targeting selective parameter updates to mitigate catastrophic forgetting while increasing computational efficiency.
- A novel merging mechanism has been developed as an extension to SWIFT to prevent parameter interference.
- Investigated privacy-preserving language adaptation of multilingual LLMs on edge devices with federated learning settings using PyTorch and Flower frameworks.
- Designed and evaluated model compression techniques (e.g. Quantization, Pruning, Distillation) and Retrieval-Augmented Generation (RAG) methods.

Research Fellow — Oct 2023 – Present
KUIS AI Center — *Istanbul, Turkey*
- Collaborated on foundational NLP research initiatives, conducting experiments on both NLP and Computer Vision

AI Researcher — Aug 2021 – Sep 2021
Baykar Defense, Computer Vision and R&D Department — *Istanbul, Turkey*
- Implemented real-time deep visual tracking and object detection models using PyTorch for autonomous unmanned aerial vehicles.

Work Experience

Senior Data Science Consultant — May 2025 – Dec 2025
Badael Company, Data & AI Team — *Riyadh, Saudi Arabia*
- Coordinated 3 cross-functional vendor teams (14 engineers) on production ML system architecture and deployment pipelines.
- Developed customer segmentation and churn propensity models to optimize operational decision-making.

Data Scientist — Nov 2021 – Jan 2025
Hepsiburada, R&D Department — *Istanbul, Turkey*
- Developed scalable customer segmentation, churn propensity, LTV prediction, and demand forecasting models.
- Built a discount sensitivity model using LightGBM and deployed automated feature and inference pipelines via Apache Airflow.
- Implemented a Two-Tower deep learning architecture to enhance personalized product recommendation systems.

Teaching Experience

Lead Teaching Assistant — Jan 2026 – Jun 2026
Modern Natural Language Processing - FAU Erlangen-Nürnberg — *Erlangen, Germany*
Assisted in course delivery, lab preparation, student project mentoring, and live PyTorch coding sessions for advanced NLP topics.

Teaching Assistant — Jan 2026 – Jun 2026
COMP 125: Introduction to Programming with Python - Koç University — *Istanbul, Turkey*
Led weekly practical instructional sessions covering fundamental Python programming and data structures.

Skills, Languages & Test Scores

Technical Skills: Python, PyTorch, HuggingFace Transformers, LangGraph, Flower, vLLM, TensorFlow, Scikit-learn, Vector Databases, Pandas, NumPy, SQL, Apache Airflow, GCP (Google Cloud Platform), BigQuery, Git
Languages: Turkish (Native), English (Advanced) | **Test Scores:** TOEFL iBT: 88, GRE Quantitative: 164